

Start With Time

Why the thing you were surest of is the floor of reality

Allen Proxmire

When I was a kid, back in the eighties, I used to take a heavy metal flashlight, turn it on in a completely dark room, and press the lens flat against my palm. You've probably done it. Your whole hand lights up a deep, glowing red, and you can see the dark shadows of the bones and the veins right through the flesh. For a second, your hand looks like a lantern.

I remember sitting there in the dark, genuinely awestruck. Because until that moment, you think of your body as a solid object — a block of stuff. And then, suddenly, you understand that it isn't solid at all. Light goes straight through it. It upends your whole baseline assumption about what's real. It felt like I'd found the first clue to a secret world. The world behind the world. The surface, it turned out, had been lying to me.

And ever since that glowing hand, I've had one lifelong obsession: what else is hiding in plain sight? If you keep pulling the thread on the fabric of reality, what's actually underneath?

This essay is about pulling that thread as far as it goes — until it reaches the one thing you were surest of your entire life, and the one thing all the clever physicists told you to distrust: time. In this one thing, your instincts are absolutely correct: time moves forward. It is not reversible or static in any sense. Much of popular science is written to tell you your gut is lying: the sun isn't moving across the sky, solid things are mostly empty space, temperature is a statistic; your senses fool you. This one is the opposite. When it comes to time — that plain feeling that time is real and things actually happen — your gut is right, and the clever people are irreversibly stuck.

Let me show you.

One second per second

Ask a physicist how fast time passes, and you'll get a knowing little smile, and the line: *one second per second*.

Consider what that means. One second, per second. It sounds like wisdom. But it's a rate of one — one over one — a number that cancels itself to nothing. As an answer to "how fast does time flow," it says exactly nothing. It's a shrug dressed up as an answer.

But that empty little *one* isn't nothing. It's a fingerprint. You only ever get a clean one when you measure something against itself. So that meaningless one is a receipt — for a decision somebody made a long time ago and then we forgot it had been made. Somewhere back in history, without announcing it, physics decided time is the same everywhere. One universal clock, ticking identically for you, for me, for a star, for a stone.

The man who signed that receipt was Isaac Newton. He laid down absolute time — a master clock ticking behind the universe, indifferent to everything, the same for everyone — and he did it so cleanly that time vanished into the machinery. Look at his law of gravity: it's two masses and the distance between them, and nothing else. There's no clock in it. Time became the stage the show plays on, and a stage is exactly the thing you stop noticing.

But there's a deep and strange thing that assumption commits you to, a piece you might not have questioned before. Snap your fingers. In Newton's universe, that exact instant is happening *everywhere at once* — not just in your room, but on the Moon, at a distant star, in the Andromeda galaxy two and a half million light-years away. A single, shared *now*, stretched across all of space, ticking in lockstep. It feels beyond questioning. Of course "now" is now, everywhere. When and where else could it be?

Albert Einstein would be the one to ask and answer just that.

The crack

The crack came from an unlikely place: the study of electricity and magnetism.

In the 1860s, James Clerk Maxwell unified them into one set of equations, and when he solved those equations for the speed of the resulting waves, the math spat out a fixed number — about 186,000 miles per second. The exact speed of light. Not a formula with variables. A hard, absolute constant.

It's easy to understate this: an absolute speed breaks the rules of everyday reality.

Speeds are supposed to be relative — a car passing you at 60 looks like 10 if you're already going 50. Maxwell's equations said light doesn't play that game. Chase a light beam at 99% of its speed, and you'd still measure it pulling away from you at the full 186,000. Turn around and flee it — still 186,000. The speed of light is the same for everyone, no matter how they move.

Newton's world and Maxwell's world could not both be true. And Einstein, a twenty-six-year-old patent clerk in 1905, outside the academic establishment, and perhaps because of it, did something audacious. He stopped trying to save Newton. He took the fixed speed of light as bedrock — one fixed fact of nature — and rebuilt everything around it.

When you do that, there is a mechanical consequence. If the speed of light must always come out the same, then other things in the equation have to bend to keep the math balanced. Distance shrinks. And time *stretches*. A clock moving really fast must run slow. Not seems to run slow — runs slow, its seconds physically longer.

This isn't a thought experiment. It has been measured in many ways. Muons — heavy, unstable particles created when cosmic rays hit the upper atmosphere — should decay in about two microseconds, far too fast to reach the ground. Yet our detectors are showered with them constantly. They survive the trip because they're moving so fast their internal clocks slow down, stretching their brief lives just enough to make it. Relativity is raining down on us from the cosmos.

So Newton's universal second died. There is no master clock. And that "shared now" we were holding? Gone. Two people moving differently don't agree on what's happening this instant across the galaxy. You and I, moving and accelerating differently, can experience the order of events differently. The present comes apart.

A mathematician named Hermann Minkowski showed that Einstein's relativity could be drawn as geometry — space and time woven into one four-dimensional fabric, space-time. And that picture gave birth to the idea that has haunted physics ever since: the *block universe*. If time is just a fourth dimension, like left or up, then all of time exists at once, every moment equally real and permanently fixed — the Big Bang at one end, the heat death at the other, and every moment of your life a static slice frozen inside. The past isn't gone; it's just behind you. The future isn't unwritten; it's just ahead of you, already there. Your sense that *now* is uniquely real, that you're making choices that shape an open future — a trick, a spotlight of consciousness sliding along a track laid down before you were born.

Physics kept dissolving time, all the way to the bottom. Try to write the single deepest equation — quantum mechanics married to gravity, the law beneath the laws — and you get something like the Wheeler–DeWitt equation, and it has a shocking feature. It has

no time in it at all. No t . Follow the erasure all the way down and you reach a description of the whole universe in which, strictly, nothing ever happens.

Most physicists took that as proof, at last, that time is not fundamental — that the flow you feel is an illusion all the way down.

To read it the other way around, we have to go back and pick up the loose thread everyone stepped over.

The arrow

Here is the most obvious fact of being alive. Time only moves forward. Drop an egg and it splatters; it never unbreaks. Coffee cools and never spontaneously reheats. You remember yesterday, never tomorrow.

Believe it or not, as written, our fundamental laws say otherwise. Write down the deepest equations for how matter behaves, and they're perfectly symmetrical — just *before* and *after*, with no rule saying which is which. Run the math backward and it's perfectly legal for a broken egg to leap off the floor and reassemble in your hand. So physics owes us an explanation. Where does the one-way arrow come from, if it's not in the math?

The textbook answer is statistics. There are countless ways to be messy and few ways to be tidy, so things slide toward disorder — toward higher entropy. But there's a gaping hole in that logic that gets glossed over. If mess is more likely than tidiness, the very same reasoning says it was more likely *yesterday, too*. Left to the odds alone, your hot coffee is a fluke that should be cooling off in *both* directions of time — hotter a minute ago being just as unlikely as hotter a minute from now. Show the statistics a half-melted ice cube and they'll tell you, with a straight face, that its most likely past is a puddle that briefly un-melted itself into a cube the instant you looked. That's insane, and we know it — an hour ago it was a solid cube from the freezer. But our equations don't know that.

So to plug the hole, physicists bolted on a separate rule: the Past Hypothesis. They simply stipulate that the universe began in a state of staggering, almost unimaginable order — a cosmic IOU, an initial condition so unlikely you couldn't write the number on a million pages. The best explanation for the arrow of time is to pin it on a fantastically fine-tuned beginning that nobody can explain.

Let's tear that IOU up and stop waiting. Let's instead just stand somewhere else.

Reality doesn't have a rewind button. Even if you could put all the atoms perfectly back in the box; all the furniture back in the same place, that configuration was preceded by

a past. The atoms, the box, the furniture, the room, and you on the couch, all commit forward. The rewind lives in the effort, never in the world. That's the clue. It's enough to consider whether instead of explaining the arrow away, to start with it. Take it as the floor.

Go all the way down — past molecules, past atoms, to the finest grain nature has. You arrive at the Planck scale, the smallest step the world can take. Down there, reality isn't a smooth substance. It's more of a process, a process in a relational substrate...

What does that mean? At the smallest scales there is not a thing, there is a *doing*. The nodes of reality are made of commitment. They are committing to realness or they are not. A series of commitments, relentless over and over commitments, make up all the objects around you. Each commitment is an update of reality and a tick of the clock. An event is made real by commitment to an outcome. The grains of reality that make up your kitchen table are screaming, "I am real", over and over. And ticking off time. Same for you, a star, a stone.

So why did every genius who wrote those equations lose track of time? Because they stepped back too far. Look at a printed photograph through a magnifying glass and you see thousands of separate ink dots. Step back and your eye blurs them into one smooth picture. Continuity is what discreteness looks like from far away. It's a trick of averaging. For example, relativity famously gives us a smooth, continuous, reversible spacetime. It's averaging over trillions of tiny one-way commitments, and an average doesn't carry an order. Of course the theory is reversible, it has no sense of before or after. A count doesn't remember its own history. We mistook the blurred average for the deep truth. The commitments — are the floor.

And that dissolves the IOU completely. If time is just committing, you don't need a miracle to explain why things run forward from an ordered start. The beginning was ordered for the plainest reason imaginable: almost nothing had been committed yet. A new notebook is clean because nothing has been written in it yet. No miracle necessary.

Start with time. Time as commitment. And this one move shows you how gravity, entanglement, and measurement come from the same place.

Gravity is uneven time

Newton gave us the math to land probes on comets. But in his own letters he admitted he had no idea what gravity actually *was* — the notion of one body reaching across millions of miles of empty vacuum to pull on another struck him as absurd. Three centuries

later, Einstein proved his discomfort was justified. There is no invisible tether. Gravity isn't a pulling force at all.

Gravity is uneven time.

Start with the fact that time really is uneven, because we've measured it. GPS satellites orbit 20,000 kilometers up, farther from Earth's mass, where the weaker gravity lets time run *faster* — about 45 millionths of a second a day faster, offset a little by their own orbital speed, for a net of roughly 38. If your phone didn't correct for the shape of time every second, your map would walk you kilometers off course by lunch. In 1971, the Hafele–Keating experiment flew atomic clocks around the world on airliners and brought them back disagreeing with the clocks left home. The Pound–Rebka experiment measured time running at different rates between the basement and roof of a single building. Your head is aging a hair faster than your feet.

So there's no single clock. The universe is a landscape where reality happens at different rates: the world ticks a little quicker up high, a little slower down deep near any mass. That isn't geometry doing something clever. It's the plain rate of time itself, thinning out as you get closer to the mass. A free thing has no engine and no plan. It just coasts, and the gradient in that ticking bends its motion toward the slower side.

Let go of your keys and nothing yanks them down. They start with no speed of their own, so the one thing they can do is follow that gradient, inward, toward the slower-ticking center. They aren't sliding down a fabric, and they aren't rolling into a dent in space. They're following the grain of time itself, and near the Earth that grain runs slower the deeper you go.

Throw a ball and it's the same grain, with one difference: now you've handed it some speed. It coasts upward into faster-ticking air, and the whole way up the gradient bends it back toward the slower side, bleeding the climb away until it stops, turns, and falls. It rises only as long as your push lasts, no farther. Nothing carries it higher, because nothing is pushing anymore. And the arc isn't the ball picking a clever route through curved space. There is no route, and no map. It's a coast through a place where time itself runs at different speeds, and the quicker-ticking air up high is what shapes the bend. Falling and arcing were never two different things. They're the same coast through the same grain of time, from two different shoves.

And this kills the most famous metaphor in science - the bowling ball on the rubber sheet. That picture puts *space* first — a pre-existing stage that mass bends. But if gravity is explained by uneven time, then the stage itself has to give: space itself isn't fundamental. It's an emergent pattern of commitment, cast like a shadow from a lamp. The world was never standing on some pre-existing stage. It's building the stage, right now,

out of commitments.

Which raises a question. If space is a shadow cast by time committing, then what is there before the shadow falls, before the substrate has committed anything at all, where distance hasn't been built yet?

No gap to cross

Those questions take us straight into the spookiest thing in physics: quantum entanglement. Take two entangled particles, separate them by light-years, measure one, and the other answers *instantly*. Einstein called it “spooky action at a distance,” and he hated it.

But it's absolutely true. The results are rock solid; a Nobel Prize was given for confirming them. No one can change the physics. But you can change the story wrapped around it.

Einstein's own fix was the sensible one: the particles must have had their answers pre-written before they parted, like a factory mailing one glove to Tokyo and one to Lima. Open the Tokyo box, see a left glove, and you instantly “know” Lima has a right — no message crossed the ocean; the answer was packed in advance. We call this “hidden variables”. Reasonable but wrong.

In 1964, John Bell found a mathematical wall: if there are hidden variables like Einstein suggested, there's a strict limit on how often the two particles can agree when you measure them in certain ways. The Nobel Prize mentioned above went to the experiments that showed the particles sail *past* Bell's wall. The matched-gloves story is dead.

But no message is sent, we know. If you measure your particle, all you see is random static — you can't tell from your side whether the other particle has even been measured, let alone what it read. The eerie matching only shows up *later*, when you physically bring the data together at ordinary speed, and compare the lists. So it's not a signal and not a pre-set secret, which leaves physics scratching its head — unless you remember what we just learned about space.

Here's the reframe. Quantum mechanics describes a pre-reality. Down at the substrate level — before a commitment and the first tick of now, the stage of space hasn't been built and distance has no meaning. Asking how far apart the two uncommitted states are is like asking what's north of the North Pole. The new picture is that there were never two separate objects somehow breaking locality. There is one scale-less pattern in the substrate, sampled twice. You measure one and the other answers instantly because

they were entangled possibilities, not committed particles in time and space yet. No message crosses the gap. There is no gap to cross until you measure. When you measure, you force a commitment, and the entangled possibilities join the universe.

The snap

Let's unpack the measurement problem because it's truly stumped physicists for a century.

Fire many single electrons at a board with two slits. Naively you would think that two spots would appear on the wall beyond. The math of quantum mechanics says no, in fact, they travel as a spread-out haze of *maybe* — a superposition — and go through both slits at once, interfering with itself to paint a striped pattern on the screen behind.

But the instant you put a detector at the slits to see which one it takes, the stripes vanish. The haze collapses. The electron picks a slit. The act of looking forces reality to choose.

This terrified the founders. The smooth equations describe the spreading haze beautifully — and go completely silent on the snap. There's no equation for the moment a wave becomes a particle; for when possibility becomes a single hard fact. And because the only thing that seemed to change was a human checking a detector, some brilliant people reached for a desperate answer: maybe the universe needs a conscious mind to look at it to make reality real.

It doesn't. A rock can collapse the haze. A speck of dust, a stray air molecule — anything an electron can interact with can force the snap. There's a reason they study quantum effects in a vacuum. And if it truly took a conscious mind, nothing would have happened in the billions of years before there was any life to look.

The mainstream, however, leans on decoherence — the well-tested fact that touching a messy (committed) environment scrambles the delicate interference and washes it away. But decoherence doesn't finish the job. It narrows the haze to a clean list of options — the electron is either here or there — and then goes silent on what actually *picks* the single answer from that list. The smooth math, even with decoherence, never actually commits.

Quantum mechanics built a universe where time is a reversible blur, a meaningless average, or just erased altogether, and then its founders were shocked they couldn't find the moment the universe makes a choice.

The snap isn't a mystery stapled onto quantum mechanics. It's commitment. Measure-

ment, entanglement, gravity — it's the universe laying down the next event. The very move that gives you the arrow of time is the move that collapses the haze, and a measurement is just one of the rare times we catch the universe in the act.

And here's where it stops being a nice story and becomes something you can *break* in a lab. Take Schrödinger's cat — supposedly alive and dead until you open the box - just as absurd as the day he penned it. But the cat is the answer, not the puzzle. The cat is a warm, wet mountain of a trillion trillion atoms. It's simply too big to become a maybe, let alone stay a maybe. The cat commits, so to speak, on its own, whether or not anyone looks. If commitment is a real physical act, it must have a size — a certain mass and internal complexity where an object gets too big to hold a superposition and is forced to commit, unobserved, perfectly isolated.

Why? Think of the standard picture: decoherence happens with an interaction with a "messy environment". A molecule, and a cat for that matter, create their own messy environment. Laboratories are hunting for this ceiling right now, coaxing molecules of thousands of atoms into the double-slit haze, climbing the mass ladder year by year. It's a real bet you can lose in a lab.

The cost

We trusted our gut and let commitment flow forward and time vary, and four towering mysteries turn into faces of a single act. The arrow, gravity, entanglement, and the snap of measurement.

No new particle. No new force. No new equation. We took the physics we already had and stood somewhere else to look at it. Whether that's a genuine unification or just a pretty story, I address honestly below.

But first, let me give you this final picture of this new universe. If commitment is the true floor, and it only runs one way, then the universe is emphatically *not* a finished film. It's not a block where past, present, and future already exist and you're strapped into a theater seat watching it play. You are sitting on the very edge of a universe being written — right now, for keeps, no undo button. You aren't watching the universe happen. You are one of the places it is happening.

That's the trade. You get a genuinely alive, open world where the future is unwritten. You give up the comforting idea that the script is already finished.

So what exactly is the framework I'm proposing? And can it be proven wrong?

Thirty-eight lines

It's called Event Density. The name is the blueprint. The world isn't built on things or static particles; it's built on *events* — the discrete happenings of the universe committing to a state. And *density* is how those happenings are distributed. A density of events is what makes a mass and so gravity. That density gradient slows local time. The name is the theory, folded into two words: how thickly the happening runs.

What I most want to show about the Event Density framework is how *little* there is to show.

Imagine you inherit an old Victorian house, every floor covered in thick, patterned carpet. Generations assumed the carpet *was* the floor and arranged their furniture around it. Then someone lifts a fraying corner and finds gorgeous original hardwood underneath. You didn't build the hardwood. You didn't import a new material. You removed an assumption that covered over the foundation for decades.

That's the whole move. For a century, the carpet was the assumption of smooth, time-reversible physics — the equations that run backward as easily as forward, that lead straight to the frozen block. Pull it up, and underneath is the discrete, one-way action of the present actually updating. That's the hardwood.

And when you pull up the carpet, the mathematics collapses to a shockingly small footprint. The whole engine fits on a single page. Thirty-eight lines.

- 13 Primitives — Axioms, not derived equations. Like space is three-dimensional, commitment is irreversible, and so on.
- 15 Postulates — Domain-specific, not derived. Such as how adjacent events couple and pass information.
- 10 Constants — Taken from measurement. The speed of light, Hubble expansion rate, Planck's constant, etc.

“Isn't importing the constants instead of predicting them just fancy bookkeeping — rearranging the same furniture and calling it a new house?” The reigning approaches — string theory is the famous case — carry a vast landscape of tunable parameters, *dials*. So for any surprising measurement, string theory can turn a dial until the math absorbs it. But a theory with enough dials can fit any result, which means it never really predicts anything.

Event Density has no dials you can turn. It needs ten numbers from Nature. The Standard Model leans on roughly two dozen. The thirteen primitives and fifteen postulates

must apply, unchanged, at every scale — the mechanics behind the arrow of time have to be identical to the mechanics behind gravity and quantum mechanics. Which makes the framework terrifyingly exposed. If a galaxy's data conflicts with the prediction, you can't isolate the problem and tweak a local dark-matter dial. One clean failed prediction, anywhere, and there's nothing to save it. A theory small enough to fit on one page, with no dials, is a theory brave enough to be demonstrably wrong. That's rare, and it's the point.

Is it true?

Science must be honest-tiered without overclaiming. Especially for a radical idea. Right now, Event Density is a *nicer story*. It unifies scattered anomalies into one narrative of temporal commitment. But the history of physics is a graveyard of nice stories. Elegance is not evidence. The universe doesn't owe us beauty.

So what makes a story more than a story? Two things.

The first is **consilience** — a word William Whewell coined in 1840. A lucky guess can fix one anomaly. But when one mechanism spontaneously resolves *four* problems that physicists treated as unrelated for decades, the odds of that being coincidence collapse toward zero. And it's stronger than that, because two of those four were enemies. Quantum mechanics and general relativity have been at war for a century. Bonus points for ending the feud.

But while consilience is compelling, it only earns a hearing. It suggests Event Density has the *shape* of truth. It doesn't prove it. What proves it is **falsifiability** — Karl Popper's line between science and story. A real theory points to a specific spot on the horizon and says: *if the data there contradicts me, I'm wrong*.

So here are three such bets, at three wildly different scales, because if one idea is really the floor, it should leave fingerprints everywhere.

Three bets

The big one — galaxies. Out at the faint edges of galaxies, stars orbit far too fast for the visible matter, as if held by unseen mass — the observation that made astronomers invent dark matter. It switches on at a specific tipping point, an acceleration scale that, for decades, has been treated as a fixed constant of nature. Event Density forbids that.

Because gravity here is tied to the expansion rate of the universe, and the universe expanded faster long ago, that tipping point *must* have been larger in the distant past and shrunk over cosmic time. The theory published that forward prediction — look at ancient, high-redshift galaxies and the number will be bigger. And in 2026, a major survey confirmed it: the tipping point is *not* a constant; it evolves, in step with cosmic expansion — and the present-day value the theory predicted landed within about 8% of what the survey measured. The long-standing assumption of a fixed constant was dismantled.

This is not a victory lap. The survey confirmed the *direction* — the number evolves — but the *rate* it evolves at came out slightly steeper than the thirty-eight lines strictly allow. Event Density is pinned, by its own logic, to track the cosmic expansion rate exactly, and the data lean a hair faster than that. It's a mild tension — roughly one to two sigma, once you fold in all the ways a single first-generation survey can be off — but it's real and unresolved, and in a zero-dial theory you can't just twist a knob to absorb it. Either those high-redshift measurements are skewed by assumptions about ancient starlight, or the coupling mechanics are genuinely off. It doesn't falsify the model, but it leaves it exposed to the next generation of telescopes. That's what a real prediction meeting real data looks like — messier than a headline, and more honest.

The middle one — a crowded gel. Take the same thirty-eight lines that govern cosmic expansion and point them at something you can hold: a protein shouldering through a dense biological gel, a solute creeping through a saturated fluid. When you map the discrete event-commitment mechanics onto a jammed local space, the result turns out to be a century-old equation for how water seeps through porous rock. It's like working out the math for a person pushing through a packed nightclub and finding it's identical to water moving through a sponge. And that derived shape fits the measured behavior of ten different crowded materials — colloids, sugar water, proteins — at very high accuracy. One honest caveat up front: what falls out of the theory blind is the *form* of the law; the exact exponents in it are still fit to each material, not predicted.

No victory lap here either. This is a *retrodiction*. The data already existed. It's a strong hint, but not a scalp. The theory is supplying a new mechanism for a fact that was already on the books. Popper demands a *forward* test, a prediction of something not yet seen, and there is one, called FRAP — fluorescence recovery after photobleaching. Take a crowded, fluorescently tagged gel, use a laser to bleach a dark circular spot, and watch the bright molecules diffuse back to heal it. Standard diffusion predicts one recovery shape. Event Density predicts a different one — the healing edge should advance at a specifically slower, nonlinear rate set by the local crowding. And this isn't a billion-dollar collider. It's a confocal microscope and a Tuesday afternoon, on kit a thousand labs already own. A guillotine sitting quietly on a lab bench, unused. If the recovery

follows standard continuous diffusion, the middle bet fails and the page is falsified.

The small one — a heavy molecule. Standard quantum mechanics puts no upper limit on superposition: isolate an object well enough — no stray photons, no gas molecules — and any mass, in principle, can stay in two places at once, forever. Event Density severs that. It draws a hard physical line: a mass ceiling where the local event substrate gets too dense to sustain the haze, and the universe is *forced* to commit — no observer, no environment required. Labs are firing ever-heavier molecules — thousands of atoms — through gratings and watching the interference that proves they took multiple paths at once. So far the superposition holds, at record masses. But the limits creep upward every few years, and that forces a fork. Either the experiments hit the predicted ceiling and the interference abruptly vanishes — an intrinsic, mass-driven commitment floor, exactly as the theory requires — or they push a molecule past it while it stays in superposition, and the whole premise dissolves. The theory dies.

Three bets. The big, the middle, the small. A lethal gauntlet — which is exactly what a theory is supposed to have.

The Growing Block Universe

The psychological toll of a century of physics has been a steady invalidation of human intuition. Relativity destroyed the universal clock. And because that shared *now* was proven false, physicists extrapolated one step further and concluded that the very *feeling* of time flowing — past behind you, future open ahead — must be an evolutionary illusion, a spotlight of consciousness sliding along a frozen block. We were conditioned to believe our lived experience of reality updating is a cognitive glitch.

But there are two different intuitions here, and everything depends on separating them. Call them the shallow gut and the deep gut. The shallow gut assumed the clock on my wall is perfectly in sync with a clock in the Andromeda galaxy — one master clock for everyone, everywhere. Relativity proved that completely wrong, and it's not coming back. But the deep gut is something else: the plain sensation that time is actively *occurring*, that the past is crystallized and finished, that the future is genuinely unwritten. And that one — if the framework is right — is exactly what Event Density hands back to you.

That's an enormous relief, honestly. You don't have to carry the intellectual burden of believing your own lived reality is a bug. The absence of a universal clock does not negate the reality of local time passing. The committing of events is the most fundamental fact there is. Your wall clock isn't syncing to a cosmic standard — but its ticking is the genuine physical update of your local corner of the world. At the scale of a human life,

on the surface of a low-gravity planet, that rate is remarkably steady. Which is exactly why Newton was right, *here*. A second is a second here in my kitchen. The passage of time is real. The present is the leading edge of a growing reality.

And this isn't some lonely outsider's conclusion. There are serious physicists, George Ellis among them, with no exposure to Event Density, who have argued for the same broad picture: the growing block universe. The universe isn't a finished crystal; it's actively accreting, adding new moments onto the permanent structure of the past. They reached that picture by a completely different road, and they'd tell it their own way, not mine. Independent roads meeting like that don't prove a theory, but they're a real point in its favor.

Meaning

If everything — a planet's orbit, a firing neuron — is just the local updating of a discrete event network, binary commitments of this-over-that, doesn't that strip all the warmth out of the world? Doesn't it make the universe a cold cosmic spreadsheet, recalculating cells?

I think it does exactly the opposite. That fear rests on a misunderstanding of what meaning *is*. We picture meaning as a warm blanket that human consciousness throws over a dead, unfeeling material world — the particles are cold, and our psychology is the comfort we drape on top. But go down to the Planck scale, below particles and fields, and there are no *things* in isolation at all. There are only relations between events. Event A couples to event B and forces the next commitment. The architecture of reality, at bottom, is pure connection and consequence.

What is meaning, when you strip it of its poetry? A memory is a relational bond across time. A promise is a commitment that dictates a future relation. A thing that matters is a thing connected to another thing. Meaning isn't a blanket over the physics. Meaning *is* relation — and relation is what the universe is made of, all the way down.

So the oldest question we have turns inside out. We spent millennia asking what the universe *means*, as if the meaning were hidden somewhere on top of it, or stamped on from outside. Event Density says the universe is *made* of meaning. Everything is connected and builds on itself.

Which is worth remembering — the next time you glance at a clock.

Everything is happening

This essay opened with a kid and a flashlight — me, a long time ago, in the eighties, pressing a glowing lens into my palm and watching the bones show through. Let me end with a different kid.

Early in my research, when there was barely any math and the idea of a universe that persistently updates was just taking root, I'd stepped away from my computer to warm up by the fireplace, still turning something over in my mind. And my daughter, four years old, trotted up to me holding her tablet, and said: *fix it. it's all happening. it's blinking.*

The phrase rang like a bell. Because it was the exact thing sitting open on my screen a few feet away, and a better picture than I had at the time. I reset her tablet. I never quite reset the thought.

I scaled her broken screen up to a billboard in Times Square once, in an earlier book. I've tried a dozen analogies since. None of them beat a four-year-old describing a glitch. Everything is happening. It's blinking.

So look around the way she did. A star is blinking — updating furiously, committing one event after another, faster than almost anything there is. A planet, a little slower. A black hole is the densest blinking there is, so many events crushed so close that the time around it slows almost to a standstill. And everything nearby leans in toward that slow, crowded knot, down the grain of time, the way the keys fell and the ball arced. That leaning is gravity.

The kettle is blinking. The page in front of you is blinking. The clock on the wall. You are blinking — every heartbeat, every thought crossing your mind right now, one update and then the next, each one set down and never taken back. All of it, everywhere, the same single act, over and over, without pause. It took a four-year-old to say it out loud.

And here's the part I keep coming back to. Not much has really changed since I was that kid with the flashlight. Same sun. Same clocks. The same physics on the board, mostly. What's changed isn't the world — it's what we're willing to see in it. My flashlight was a bulb and two heavy batteries. Hers is a screen smaller and more powerful than anything that existed in 1980, and she pointed it at reality and said *it's all happening* like it was the most obvious thing in the world. For me, that took a flashlight, and a lifetime, and a book. Maybe for her it just starts out obvious.

I'll end here with one last thought: the first and last blinks. Think about what a blink even is. It's a change. A difference. A *this* where a moment before there was a *that*. A difference where before there was nothing to tell apart at all. A blink is a contrast. So

the first blink would have to be the first *difference*: the first time a perfectly smooth, uniform universe made a difference. The last blink would have to be the *last difference to have meaning*. The first and last this-and-not-that.

What made the very first difference, in a universe that started without any? I don't know.

But it happened, and it's all happening.